

Step 1: Configure Development and Production Profiles in Spring Boot

AWS Lightsail Deployment Course

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Table of contents

1	Configure Development and Production Profiles	2
2	Why Spring Boot Profiles Matter	2
3	Base Configuration Setup	3
4	Development Profile Configuration	3
5	Production Profile Configuration	4
6	Environment Variables in Production	4
7	Build Configuration Updates	5
7.1	Maven Dependencies	5
8	Testing Your Profiles	6
8.1	Development Profile	6
8.2	Production Profile Testing	6
9	Building for Production	6
10	Configuration Verification Checklist	7
11	Key Benefits Achieved	7
12	What's Next?	8
13	Summary	8

1 Configure Development and Production Profiles

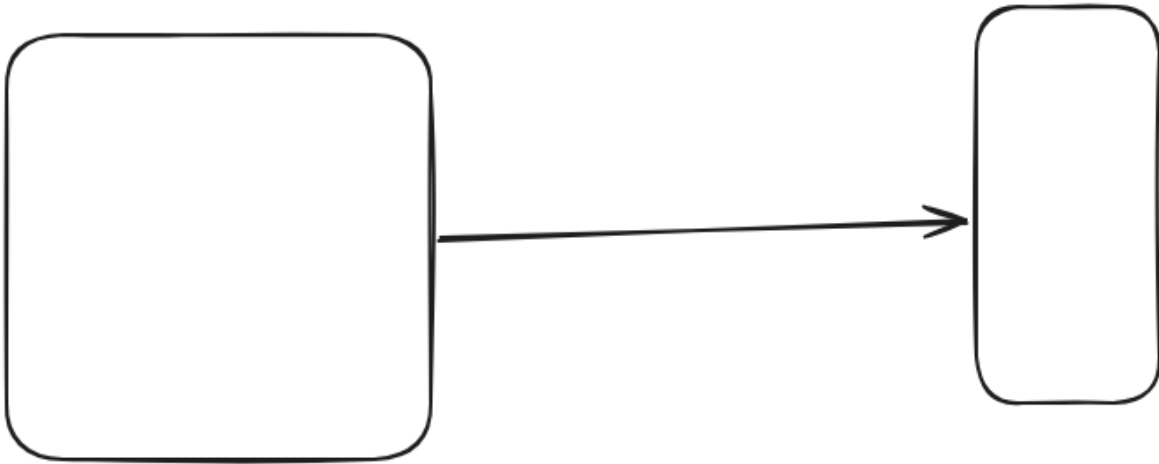


Figure 1: Preparing Spring Boot for AWS Lightsail

Preparing your Spring Boot application for seamless deployment to AWS Lightsail

Welcome to step 1 of deploying Spring Boot apps to AWS Lightsail. Today we'll configure separate application profiles for development and production environments.

2 Why Spring Boot Profiles Matter

Development Environment

- Local database connections
- Debug logging enabled
- Development-friendly settings

Production Environment

- Cloud database connections
- Optimised performance settings
- Security-focused configuration

Profiles allow you to segregate application configuration and make it available only in certain environments. This prevents hardcoding production credentials and enables local testing with development databases.

3 Base Configuration Setup

Create `application.properties` with common settings:

```
# Application name and basic settings
spring.application.name=my-spring-app
server.servlet.context-path=/

# Default active profile
spring.profiles.active=dev

# JPA/Hibernate common settings
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=false
```

The base configuration contains settings that apply to all environments. Notice we set the default profile to 'dev' for local development convenience.

4 Development Profile Configuration

Create `application-dev.properties`:

```
# Local MySQL database configuration
spring.datasource.url=jdbc:mysql://localhost:3306/myapp_dev
spring.datasource.username=dev_user
spring.datasource.password=dev_password

# Development-specific settings
spring.jpa.show-sql=true
logging.level.org.springframework.web=DEBUG
server.port=8080
```

The development profile uses local database connections and enables verbose logging to help with debugging during development.

5 Production Profile Configuration

Create `application-prod.properties`:

```
# Production database with environment variables
spring.datasource.url=${DB_URL:jdbc:mysql://localhost:3306/myapp_prod}
spring.datasource.username=${DB_USERNAME:prod_user}
spring.datasource.password=${DB_PASSWORD:secure_password}

# Production optimizations
spring.jpa.show-sql=false
spring.jpa.hibernate.ddl-auto=validate
server.compression.enabled=true
```

Production profile uses environment variables for database configuration, enabling secure credential management without hardcoding sensitive information.

6 Environment Variables in Production

Syntax: `${VARIABLE_NAME:default_value}`

Benefits:

- Secure credential management
- Environment-specific overrides
- No hardcoded production secrets
- Easy configuration changes

```
spring.datasource.url=
    ${DB_URL:default}

spring.datasource.username=
    ${DB_USERNAME:user}

spring.datasource.password=
    ${DB_PASSWORD:pass}
```

The `${}` syntax allows overriding configuration values with environment variables while providing fallback defaults. This is crucial for cloud deployments.

7 Build Configuration Updates

7.1 Maven Dependencies

```
<dependency>
  <groupId>mysql</groupId>
  <artifactId>mysql-connector-java</artifactId>
  <scope>runtime</scope>
</dependency>

<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
```

Ensure your build configuration includes the necessary dependencies for MySQL connectivity and JPA data access.

8 Testing Your Profiles

8.1 Development Profile

```
# Run with development profile (default)
mvn spring-boot:run

# Or explicitly specify
mvn spring-boot:run -Dspring-boot.run.profiles=dev
```

8.2 Production Profile Testing

```
# Set environment variables
export DB_URL="jdbc:mysql://localhost:3306/myapp_test"
export DB_USERNAME="test_user"

# Run with production profile
mvn spring-boot:run -Dspring-boot.run.profiles=prod
```

Testing both profiles locally ensures your configuration works correctly before deployment. The production profile test helps verify environment variable substitution works.

9 Building for Production

Create a deployment-ready JAR:

```
# Clean and build the project
mvn clean package

# Verify the JAR was created
ls -la target/*.jar
```

Output:

```
-rw-r--r-- 1 user user 45M Dec  1 10:30 my-spring-app-1.0.0.jar
```

Why a JAR?

The executable JAR bundles your application and all dependencies — one file ready for cloud transfer.

The executable JAR contains your application and all dependencies, making it perfect for cloud deployment. This single file can be transferred to your Lightsail instance.

10 Configuration Verification Checklist

- Created `application.properties`, `application-dev.properties`, and `application-prod.properties`
- Application starts successfully with development profile
- Executable JAR builds without errors
- Database dependencies configured in build file
- Environment variable syntax working in production profile

Before proceeding to the next step, ensure all these checkpoints are completed. This foundation is crucial for successful deployment.

11 Key Benefits Achieved

Development Benefits

- Local development database
- Debug logging enabled
- Fast development cycle
- Easy configuration changes

Production Benefits

- Secure credential management
- Optimised performance settings
- Environment-specific configurations
- Cloud-ready deployment package

This profile configuration provides the foundation for a professional deployment workflow, separating concerns between development and production environments.

12 What's Next?

Coming Up in Step 2

- Create AWS Lightsail instance
- Configure Linux environment
- Set up MySQL database
- Prepare for application deployment

Your Spring Boot application is now **production-ready** with proper profile configuration!

With profiles configured, your application can seamlessly transition from local development to AWS Lightsail. Next, we'll create the cloud infrastructure to host your application.

13 Summary

- **Spring Boot profiles** enable environment-specific configurations
- **Environment variables** provide secure credential management
- **Executable JARs** simplify deployment processes
- **Proper configuration** is essential for cloud deployments

Your application is now ready for AWS Lightsail deployment!

You've successfully configured a Spring Boot application with separate development and production profiles, created a deployment-ready JAR, and established the foundation for secure cloud deployment.